

Cryptography Notes

Oli

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1. Algebra

1.1. Notation

- Z = 整數 integers
- $Z_n = \{0, 1, 2, \dots, n-1\}$ ($0 \sim n-1$ 的整數)
- $Z_n^* = \{a \in Z_n \mid \gcd(a, n) = 1\}$ (Z_n 裡面和 n 互質的)

eg. $Z_{12}^* = \{1, 5, 7, 11\}$

- N = non-negative integers (有 0)
- P = positive integers (無 0)
- Q = rational numbers
- R = real numbers
- C = complex numbers
- Q^*, R^*, C^* = **non-zero** rational, real, complex numbers
- $GL_n(R)$
 - $n \times n$ 的可逆 (invertible) 方陣
 - 裡面的元素 (entries) 都在 R 裡
- $SL_n(R)$
 - $n \times n$ det 為 1 的方陣 (det 為 1 所以也可逆)
 - 裡面的元素都在 R 裡、是 $GL_n(R)$ 的 subset

1.2. Group

1.2.1. Definition

group $(G, *)$ 是一個集合 G 跟一個運算 $*$ 所構成的 (中文是群)

一個 group 要滿足四個條件：

1. Closure 封閉性

$$a * b \in G \quad \forall a, b \in G$$

$a * b$ 後的答案還是得在 G 裡面

2. Associativity 結合律

$$a * (b * c) = (a * b) * c \quad \forall a, b, c \in G$$

3. Identity

$$\exists e \in G \text{ s.t. } a = a * e = e * a$$

找得到一個 identity e 讓 a 和 e 運算或 e 和 a 運算後還是自己

* 通常是加或乘

加法的 identity e 通常是 0 (0 加自己或自己加 0 都還是自己)

減法的 identity e 通常是 1 (1 乘自己或自己乘 1 都還是自己)

4. Inverse 反元素

$$\forall a \in G, \exists b \in G \text{ s.t. } a * b = b * a = 1$$

找得到跟和 a 運算完會變 identity 的 inverse b

Group 的 Example

- Z, Q, R, C with $+$
- Q^*, R^*, C^* with $\times$
- $5Z = \{5a \mid a \in Z\}$ with $+$
- $\{1, -1\}$ with $\times$
- $Z_6 = \{0, 1, 2, 3, 4, 5\}$ with $+\text{mod } 6$
- Z_7^* with $\otimes$ ($\otimes$ 表示 $\times \text{mod } p$, p 是 Z 的下標 所以在這邊 $\otimes = \times \text{mod } 7$)
- $\{(x, y) \in R^2 \mid y^2 = x^3 + ax + b\} \cup \{\infty\}$ with point addition (加法運算：兩個點相加) and point doubling (兩倍運算：一個點的兩倍) on elliptic curves

不符合 Group 的 Example

- Z_7 with $+$ $\Rightarrow$ not closed
- P with $+$ $\Rightarrow$ no identity
- Z with $-$ $\Rightarrow$ no associativity
- $2Z + 1 = \{2a + 1 \mid a \in Z\}$ (奇數) with $\times \Rightarrow$ no inverse

1.2.2. Proposition

1. The identity e of G is unique

Proof

假設有兩個 identity e_1 跟 $e_2 \Rightarrow e_1 = e_1 * e_2 = e_2$

2. The inverse b of each $a \in G$ are unique, denoted as a^{-1}

Proof

假設 b 跟 c 都是 a 的 inverse $\Rightarrow b = b * e = b * (a * c) = (b * a) * c = e * c = c$

3. $(a^{-1})^{-1} = a \forall a \in G$
4. $(a * b)^{-1} = b^{-1} * a^{-1} \forall a, b \in G$
5. $\forall a, b, c \in G$
 - a. $a * x = b$ and $y * a = b$ have unique solution

Proof

$a^{-1} * (a * x) = a^{-1} * b \rightarrow (a^{-1} * a) * x = a^{-1} * b \rightarrow e * x = a^{-1} * b$
 $\Rightarrow x = a^{-1} * b$

- b. $a * b = a * c$ 代表 $b = c$ and $a * b = c * b$ 代表 $a = c$

Proof

兩邊同乘 a^{-1} 然後套一下四個性質

1.2.3. Permutation

A **permutation** is a **bijective** function $f : T \rightarrow T$ on a set T into itself

injective 單射 (one-to-one)	surjective 滿射 (onto)	bijective 雙射 (one-to-one and onto)

Permutation 的 Example

$T = \{1, 2, 3\}$ with $f(1) = 2, f(2) = 3, f(3) = 1$ is a permutation represented by the array

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

就是矩陣裡面上面那排的經過 f 函數會被換成下面那排的

Remark

Every block cipher with a specific key is a permutation

- $DES_k : \{0, 1\}^{64} \rightarrow \{0, 1\}^{64}$ (block size = 64 bits)
- $AES_k : \{0, 1\}^{128} \rightarrow \{0, 1\}^{128}$ (block size = 128 bits)

1.2.4. Symmetric Group

A **symmetric group** S_n is the set of all permutations on $\{1, 2, 3, \dots, n\}$ with $\circ$ (composition)

$(f \circ g)(x) = f(g(x))$ (合成函數 先運算後面的再運算前面的)

Symmetric Group 的 Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \circ \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}$$

$$S_3 = \left\{ \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix} \right\}$$

可以把上面那些東西簡寫 : $S_3 = \{e, (12), (13), (23), (123), (132)\}$

- $(123)^{-1} = (321) = (132)$

Remark

The S-box of AES is a permutation in S_{256}

1.2.5. Abelian Group

A group $(G, *)$ is **commutative** or **abelian** if $a * b = b * a \forall a, b \in G$

Abelian Group 的 Example

- Z, Q, R, C with $+$
- $Z_9^* = \{1, 2, 4, 5, 7, 8\}$ with $\otimes$
在 Z_9^* 中 2 和 5 的次方都會跑遍所有元素 所以他們是 Z_9^* 的生成元 (generator)
以 2 為例: $1 = 2^6, 2 = 2^1, 4 = 2^2, 5 = 2^5, 7 = 2^4, 8 = 2^3$
- Z_p^* with $\otimes$ for every prime p

Non-Abelian Group 的 Example

只要找到任意一個 $a * b \neq b * a$ 就是了

- for $n \geq 3$, the symmetric group S_n is not abelian :

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix} \circ \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix} \neq \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \circ \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}$$

- with the operation of matrix multiplication, each of the following is not abelian :
 - ▶ A general linear group $GL_n(R)$
 - ▶ A special linear group $SL_n(R)$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

the above is in $SL_2(Q) \subset GL_2(Q)$

1.2.6. Cyclic Group

A group $(G, *)$ is **cyclic** if there exists a generator $g \in G$ s.t. every $a \in G$ is of the form $a = g * g * g * \dots * g$ (n copies) for some $n \in Z$ (n 可以 < 0)

- 2 copies $\Rightarrow a = g * g$
- -2 copies $\Rightarrow a = g^{-1} * g^{-1}$

Cyclic Group 的 Example

- $(Z, +)$ with generators 1 and -1
- $(Z_7^*, \otimes)$ with generator 3
- $(Z_9^*, \otimes)$ with generators 2 and 5

不是 Cyclic Group 的 Example

- $(Q, +)$
- $(Z_8^*, \otimes)$ (這種有 4 個元素而且都不是 generator 的 group 叫 Klien 4)

1.2.7. Group Order

The **order** of a group $(G, *)$ (denoted as $|G|$ or $\#G$) is the number of the elements in G , or the **cardinality** of G

Group Order 的 Example

- $|S_3| = 6, |S_4| = 24, |S_n| = n!$
- $|Z_p| = p, |Z_p^*| = p - 1$ for any prime p
- $|Z_9^*| = 6$
- $|Z_n^*| = |\{a \in Z_n \mid \gcd(a, n) = 1\}| = \phi(n)$
 $\phi(n)$ 叫 euler phi-function
- 用來 construct AES 的 S-box 的 $|GL_8(Z_2)| = ??$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \in GL_8(Z_2)(= SL_8(Z_2))$$

A group $(G, *)$ is **finite** if $|G| < \infty$, and is **infinite** if $|G| = \infty$

Finite / Infinite 的 Example

- $|Z| = \infty, |Q| = \infty, |R| = \infty, |C| = \infty$
 - $|Z|$ and $|Q|$ have the same cardinality, which is countable
 - $|R|$ and $|C|$ have the same cardinality, which is uncountable

1.2.8. Subgroup and Coset

A subset H of a group $(G, *)$ is a **subgroup** of G if H is itself a group under $*$ in G

A subgroup H of a group $(G, *)$ is a subgroup of G iff

1. $a * b \in H \forall a, b \in H$
2. $a^{-1} \in H \forall a \in H$

Subgroup 的 Example

- $(5Z, +)$ is a subgroup of $(Z, +)$
- $(Q^*, \times)$ is a subgroup of $(R^*, \times)$

If H is a subgroup of a group G , and $g \in G$, then

- $gH := \{g * h \mid h \in H\}$ is a **left coset of H** in G
- $Hg := \{h * g \mid h \in H\}$ is a **right coset of H** in G
- for abelian groups, $gH = Hg$

Remark

In general, Left coset $\neq$ Right coset

☰ Coset 的 Example

- $(123)\{e, (12)\} = \{(123), (13)\}$ ((123) 是 S_3 的元素、 $\{e, (12)\}$ 是 S_3 的 subgroup)
 $\{e, (12)\}(123) = \{(123), (23)\}$
 $\Rightarrow L \neq R$
- $(12)\{e, (123), (132)\} = \{(12), (23), (13)\}$
 $\{e, (123), (132)\}(12) = \{(12), (13), (23)\}$
 $\Rightarrow L = R$

$$g_1 H = g_2 H \Leftrightarrow g_1^{-1} g_2 \in H$$

☰ 這個性質的 Example

$$(123)\{e, (12)\} = (13)\{e, (12)\} \Leftrightarrow (123)^{-1}(13) = (132)(13) = (12) \in \{e, (12)\}$$

1.2.9. Lagrange Theorem

If H is a subgroup of a finite group G , then $|H|$ divides $|G|$

H is a subgroup of a finite group G , then $[G : H] := \frac{|G|}{|H|}$ is the **index** of H in G

For $g \in G$, the **order** of g , denoted by $o(g)$, is the smallest positive integer n s.t. $g^n = e$
 $\rightarrow o(g)$ divides $|G| : \langle g \rangle = \{g, g^2, g^3, \dots, g^{o(g)-1}, g^{o(g)} (= e)\}$ is a subgroup of G

☰ 這個性質的 Example

- $\{e, (123), (132)\}$ is a subgroup of S_3 , $3 \mid 6$
- $\{1, 8\}$ is a subgroup of $(Z_9^*, \otimes)$, $2 \mid 6$
- $SL_2(Z_7)$ is a subgroup of $GL_2(Z_7)$
 $|GL_2(Z_7)| = (7^2 - 1)(7^2 - 7)$, $|SL_2(Z_7)| = \frac{|GL_2(Z_7)|}{6}$ (下面有解釋)
- $T_2(Z_7) = \{M \in GL_2(Z_7) \mid M \text{ is upper triangular}\}$ is a subgroup of $GL_2(Z_7)$
 $|T_2(Z_7)| = 6 \times 6 \times 7$

$|SL_2(Z_7)| = \frac{|GL_2(Z_7)|}{6}$ 的解釋 (可以推廣至 $|SL_n(Z_p)| \cdot |GL_n(Z_p)|$)

1. Let $H = SL_2(Z_7) = \left\{ \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \mid a_{ij} \in Z_7, a_{11}a_{22} - a_{12}a_{21} = 1 \right\}$
2. For $t = 2, 3, 4, 5, 6$, define the left coset $H_t = \begin{bmatrix} 1 & 0 \\ 0 & t \end{bmatrix} H$, then
 - a. $H_t = \{M \in GL_2(Z_7) \mid \det(M) = t\}$
 - b. H and H_t 's are mutually disjoint (兩兩互斥：任兩個集合中沒有共同元素)
 - c. $f_t : H \rightarrow H_t$ defined by $f_t(M) = \begin{bmatrix} 1 & 0 \\ 0 & t \end{bmatrix} M$ is bijective
 - d. $|H| = |H_t|$
 - e. $GL_2(Z_7) = H \cup H_2 \cup H_3 \cup H_4 \cup H_5 \cup H_6$
3. $|GL_2(Z_7)| = |H| + |H_2| + |H_3| + |H_4| + |H_5| + |H_6| = 6 \times |SL_2(Z_7)|$

1.2.10. Homomorphism and Isomorphism

$(G, *)$ and $(H, \star)$ are two groups

- A function $f : G \rightarrow H$ is a **homomorphism** (同態) if
 - $f(a * b) = f(a) \star f(b) \forall a, b \in G$
- A function $f : G \rightarrow H$ is a **isomorphism** (同構) (G is isomorphic to $H / G \cong H$) if
 - $f(a * b) = f(a) \star f(b) \forall a, b \in G$
 - f is bijective

Homomorphism 的 Example

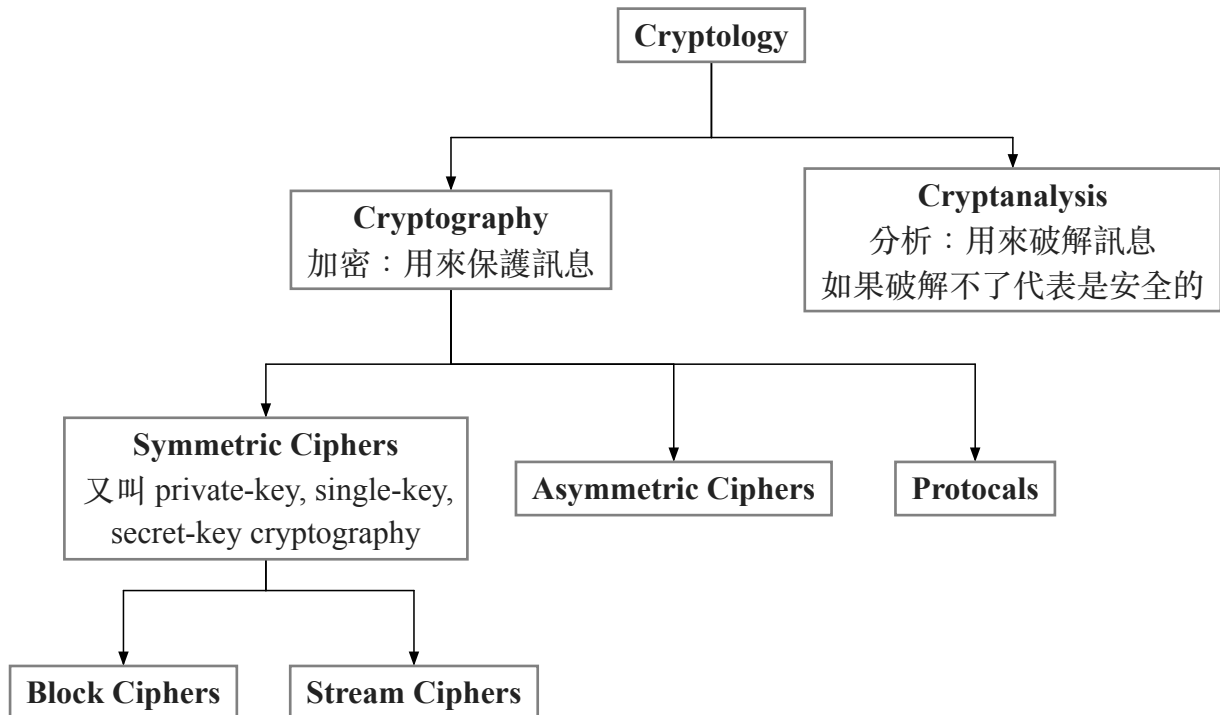
- $f : (Z, +) \rightarrow (Z, +)$ is a homomorphism with $f(1) = 2$
 - injective (one-to-one), not surjective (onto)
- $f : (Z, +) \rightarrow (Z_7^*, \otimes)$ is a homomorphism with $f(1) = 3$
 - surjective (onto), not injective (one-to-one)

Isomorphism 的 Example

- $f : (C^*, \times) \rightarrow (C^*, \times)$ defined by $f(a + bi) = a - bi$
- $f : (Z_6, + \bmod 6) \rightarrow (Z_7^*, \times \bmod 7)$ is a isomorphism defined with $f(1) = 3$
 - $f(2) = f(1 + 1) = 2, f(3) = f(1 + 1 + 1) = 6, f(4) = 4, f(5) = 5, f(0) = 1$
 - $f(0) = 1$ 因為 $f(0) = f(0 + 0) = f(0) \times f(0)$ 兩邊同乘 $f(0)^{-1}$ 得到 $f(0) = 1$
 - 思考：如果 $f(1) = 5, f(1) = 2$ 會怎麼樣

2. Cryptography

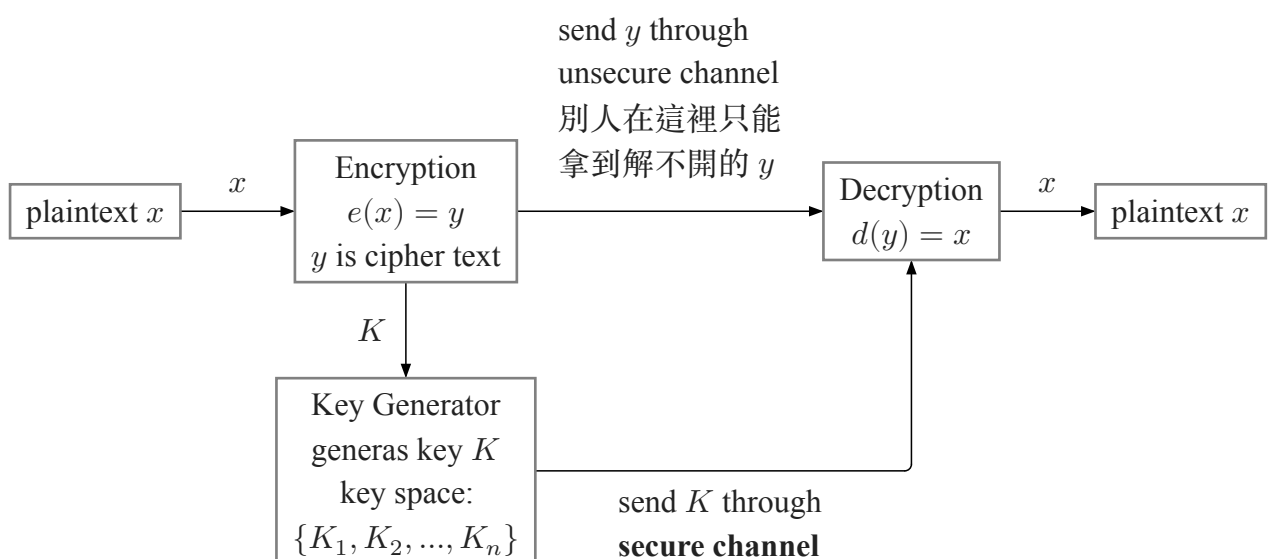
2.1. Introduction



Basic Facts

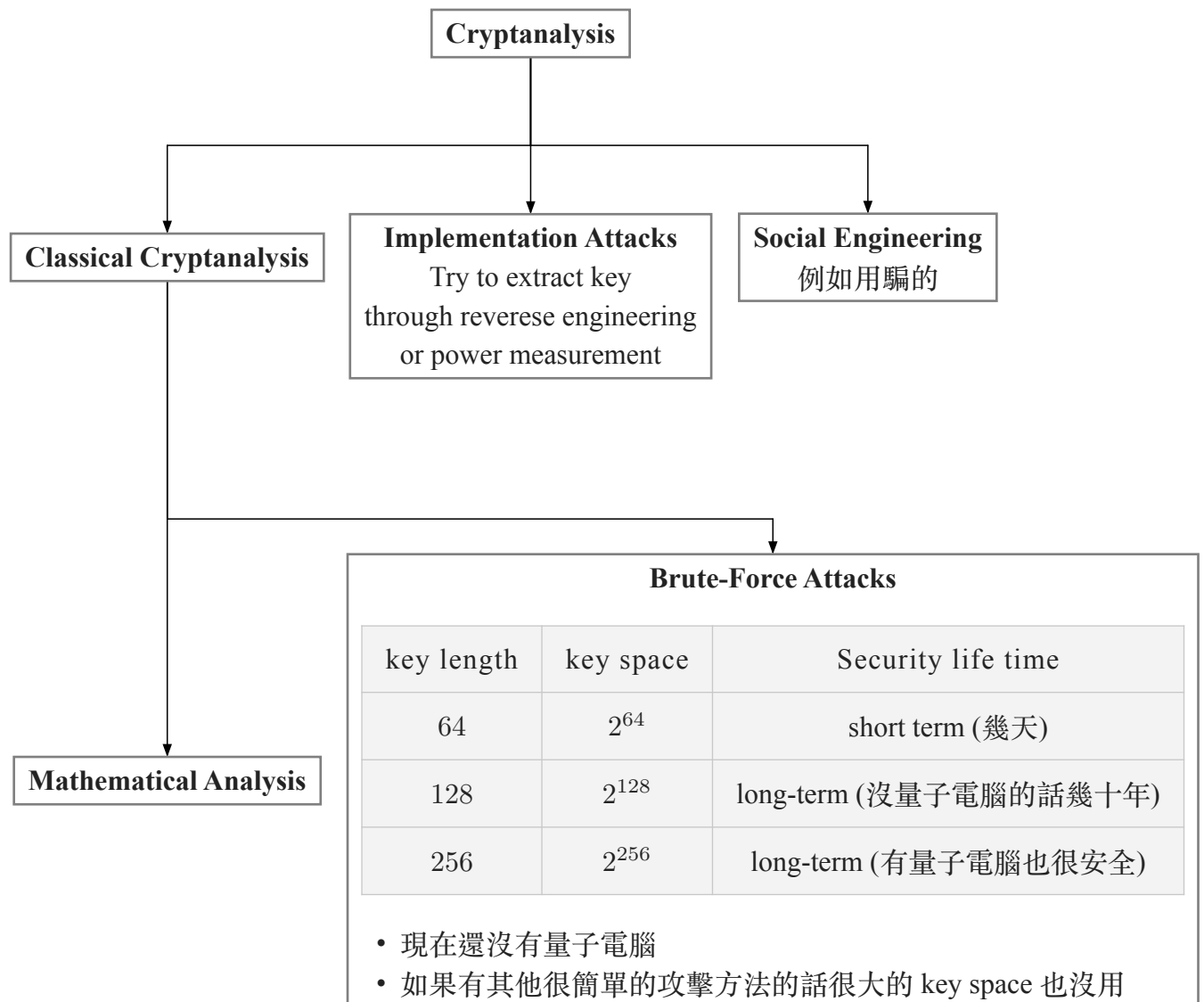
- Ancient Crypto: 2000 B.C. 古埃及 Letter-based encryption scheme 就很紅 (Ceasar cipher)
- Symmetric Ciphers: 從古時候到 1976 的加密都是 Symmetric 的
- Asymmetric Ciphers: 1976 Diffie, Hellman, Merkle 發明了 public-key cryptography
- Hybrid Schemes: 現在用的 protocol 大部分是都有用的
 - symmetric ciphers for encryption and message authentication
 - asymmetric ciphers for key exchange and digital signature

2.1.1. Symmetric Cryptography



encryption equation and decryption equation are inverse operations: $d_K(y) = d_K(e_K(x)) = x$

2.1.2. Cryptanalysis



2.1.3. Substitution Cipher

方法：把每個字母都選一個特定的字母換掉
有兩種解法：

Exhaustive Key Search (Brute-Force)	Letter Frequency Analysis (Brute-Force)
一個一個找 有 $26 \times 25 \times \dots \times 3 \times 2 \times 1 = 2^{88}$ keys substitution table (=keys) 超大	出現頻率最高的英文字母是 e (大概 13%) 方法： 1. 用 e 換掉在 cipher text 裡面出現頻率最高的字母 2. 再用頻率次高的字母換掉 cipher text 裡面出現第二多次的字母 → 要試的次數少很多

2.1.4. Shift (Caesar) Cipher and Affine Cipher

兩種方法都會用把字母對應到數字 $\rightarrow A = 0, B = 1, \dots, Z = 25$

Shift (Caesar) Cipher

把每個字母用它 $+ \text{key} \bmod 26$ 換掉

Let $k, x, y \in \{0, 1, \dots, 25\}$

- Encryption: $y = e_k(x) \equiv x + k \bmod 26$
- Decryption: $x = d_k(x) \equiv y - k \bmod 26$

key space is only 26

Affine Cipher

比 Shift Cipher 更高級一點

key : (a, b)

Let $a, b, x, y \in \{0, 1, \dots, 25\}$

- Encryption: $y = e_k(x) \equiv ax + b \bmod 26$
- Decryption: $x = d_k(x) \equiv a^{-1}(y - b) \bmod 26$

a 只能選 $\gcd(a, 26) = 1$ 的 不然沒有 inverse (總共有 12 個)

key space is $12 \times 26 = 312$